

SRI VASAVI ENGINEERING COLLEGE (Autonomous)
B.Tech V Semester Regular Examinations
Natural Language Processing
 (AIML)

Time: 3 Hrs

Max. Marks: 70

PART-A Answer All the Questions.			
1.			20 M
	a	Define Statistical Language Modelling and give one example	CO1-K1(2M)
	b	Write the purpose of Regular Expressions in NLP	CO1-K1(2M)
	c	What is the role of smoothing in N-gram models?	CO2-K1(2M)
	d	Differentiate between Rule-based and Stochastic POS tagging.	CO2-K1(2M)
	e	List the main steps in Shallow Parsing.	CO3-K1(2M)
	f	List any two advantages of Dependency Grammar	CO3-K1(2M)
	h	Define Semantic Attachment with an example	CO4-K1(2M)
	g	What is Word Sense Disambiguation? Give one practical example.	CO4-K1(2M)
	h	Mention any two applications of WordNet in NLP.	CO5-K1(2M)
	i	What is PropBank in NLP?	CO5-K1(2M)
PART-B All Questions Carry Equal Marks			
2.			10 M
	A.	i. Evaluate the challenges NLP systems face in handling ambiguity and multilingual contexts. Support your answer with examples	CO1-K2
		Describe the process of Tokenization with examples.	CO1-K2
		OR	
	B.	i. Create a Finite-State Automaton (FSA) that accepts the strings: "cat", "bat", "rat", and rejects "cot", "bit", "run". Represent it using a state diagram.	CO1-K2
		Compute the Levenshtein distance between the words "kitten" and "sitting".	CO1-K2
3.			10 M
	A.	i. Contrast Unsmoothed and Smoothed N-gram models with examples.	CO2-K3
		ii. Solve for the given the corpus below, compute the bigram probability P ("is raining" "it"). Corpus: "it is raining, it is sunny, it was cold, it is raining heavily."	CO2-K3
		OR	
	B.	i. Using a given corpus, calculate the unigram and bigram probabilities for the sentence: "I love NLP". Apply Laplace smoothing.	CO2-K3
		ii. Construct Hidden Markov Model for POS tagging with transition and emission probabilities	CO2-K3
4.			10 M
	A.	i. Explain Context-Free Grammar rules for English sentence construction.	CO3-K2
		ii. Demonstrate CYK parsing for the sentence "The cat sleeps" using a given grammar.	CO3-K3
		OR	
	B.	i. Discuss various types of ambiguities in syntactic parsing	CO3-K2
		ii. Investigate on Feature Structures and Unification in NLP.	CO3-K3
5.			10 M
	A.	i. Explain the process of Word Sense Disambiguation using Supervised Learning	CO4-K2
		ii. Using Lesk's algorithm, disambiguate the word 'bank' in 'He went to the bank to deposit money'	CO4-K3
		OR	
	B.	i. Describe Thematic Roles with suitable examples	CO4-K2
		ii. Illustrate the steps for Bootstrapping method of WSD.	CO4-K3
6.			10 M
	A.	i. Enumerate on Discourse Segmentation and Coherence with examples.	CO5-K2

		ii.	Distinguish on PropBank and FrameNet in detail	CO5-K3
			OR	
	B.	i.	Explain the structure and applications of British National Corpus.	CO5-K2
		ii.	In 'John went to the park. He played football there'. Identify antecedent for 'He' using Hobbs algorithm.	CO5-K3
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